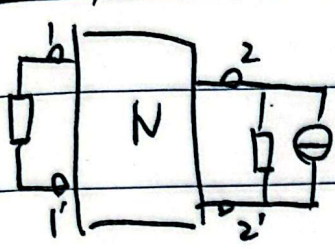
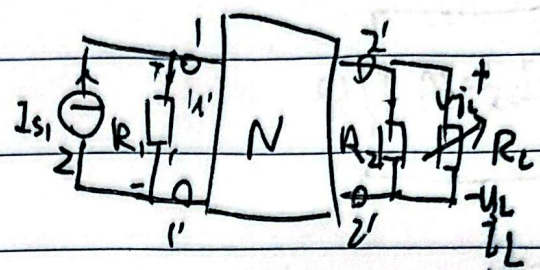


4.43



$$\left. \begin{aligned} u_1 &= 1 \\ i_1 &= 1 \\ u_2 &= 2 \\ i_2 &= -2 \end{aligned} \right\} (SZ)$$



$$\left. \begin{aligned} \hat{u}_1 &= u' \\ \hat{i}_1 &= i_1' \\ \hat{u}_2 &= u_L \\ \hat{i}_2 &= i_L + \frac{u_L}{2} \end{aligned} \right\}$$

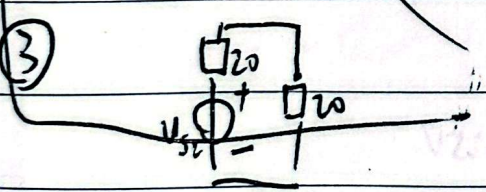
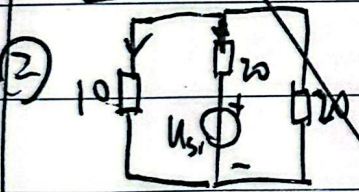
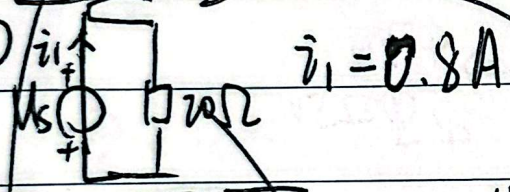
$$\Rightarrow \cancel{2u'} + 2i_L + u_L = u' - 2u_L$$

$$\Rightarrow 2i_L + 3u_L = 2$$

$$P_{uL} = u_L i_L = \cancel{u_L} \cdot \frac{2-2i_L}{3} = \frac{2}{3}(1-i_L)i_L \quad \text{当 } i_L = \frac{1}{2}A, P_{\max} = \frac{1}{6}W$$

4.46

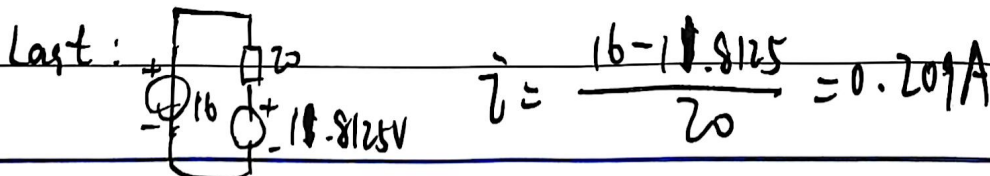
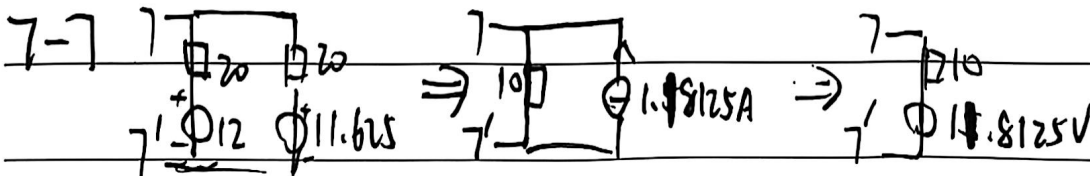
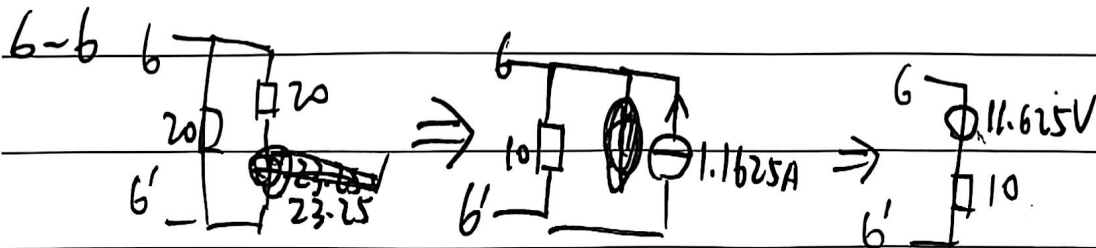
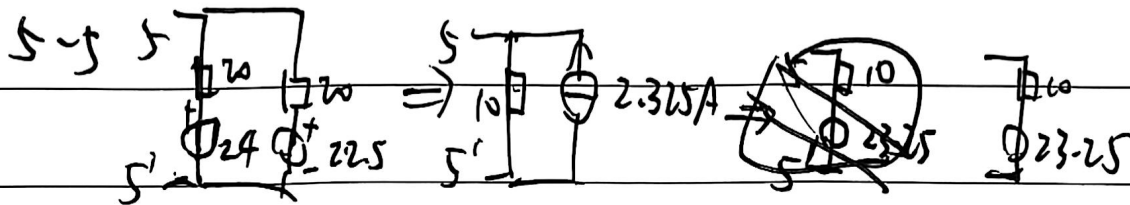
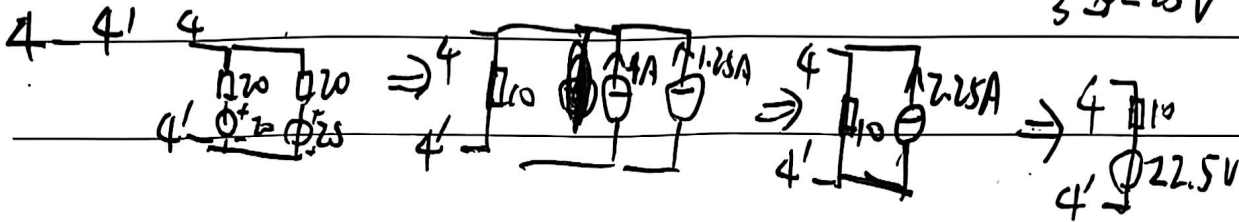
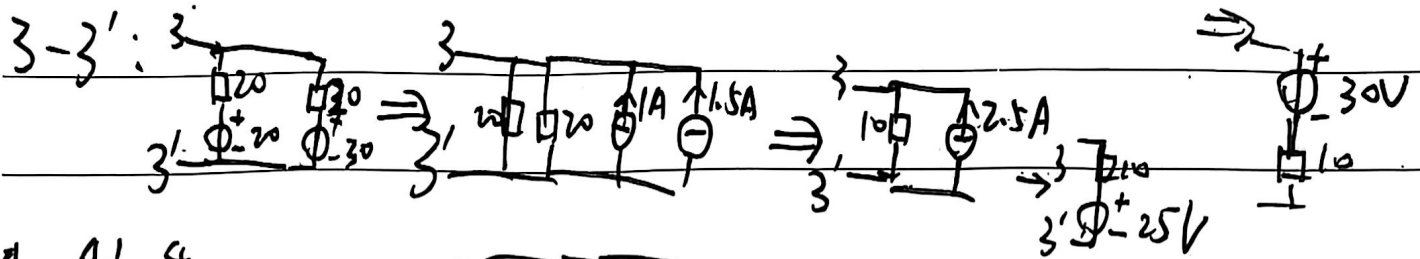
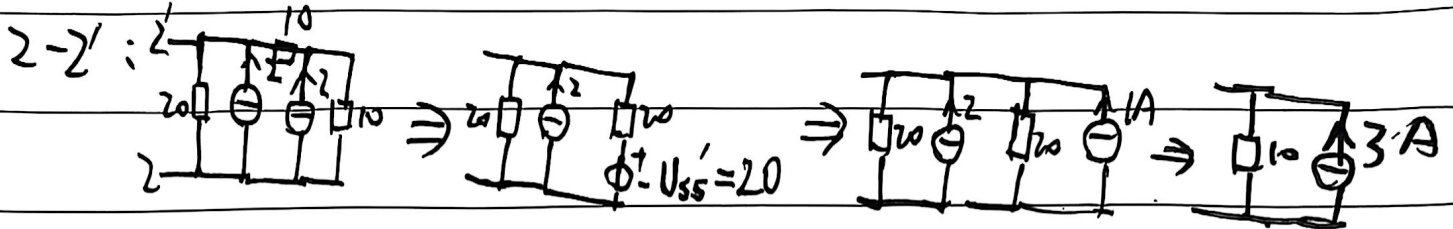
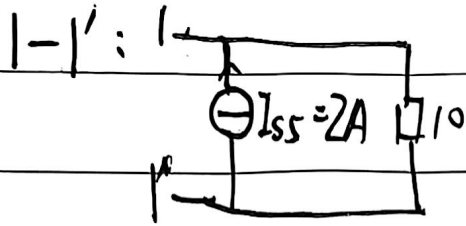
叠加:

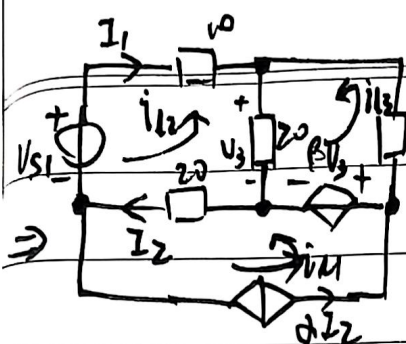


$$i_{20} = \frac{u_{s1}}{\frac{80}{3}\Omega} = 0.45A \quad i_2 = \frac{1}{10} \left(\frac{1}{4} u_{s1} \right) = 0.3A$$

4.46

or ~~叠加~~ + ~~等效~~ ~~互易~~:





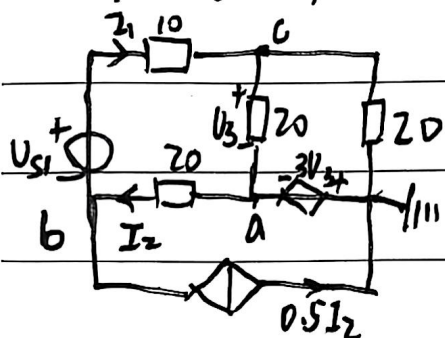
① 回路电流法:

$$\begin{cases} i_{L1} = 0.5 I_2 = 0.5 i_{L2} \\ 50 i_{L2} + 60 - 20 i_{L1} - 20 i_{L3} = 0 \\ 40 i_{L3} - 3 U_3 - 20 i_{L2} = 0 \\ I_2 = i_{L1} - i_{L2} \\ U_3 = (i_{L3} - i_{L2}) \cdot 20 \end{cases}$$

$$\Rightarrow \begin{cases} i_{L2} = -2A \\ i_{L1} = 2A \\ i_{L3} = -4A \end{cases} \Rightarrow I_1 = -i_{L2} = 2A$$

$$\therefore P = V_{S1} I_1 = 120W \text{ 发出功率 } 120W$$

② 节点电压法

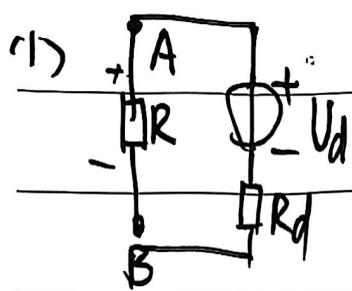


$$\begin{cases} U_a = -3U_3 \\ \frac{U_b - U_c}{10} + \frac{U_b - U_a}{20} = -\frac{V_{S1}}{10} - 0.5 I_2 \\ \frac{U_c}{20} + \frac{U_c - U_a}{20} + \frac{U_c - U_b}{10} = \frac{V_{S1}}{10} \\ I_2 = \frac{U_a - U_b}{20} \\ U_3 = U_c - U_a \end{cases}$$

$$\Rightarrow \begin{cases} U_a = 120V \\ U_b = 40V \\ U_c = 80V \end{cases} \Rightarrow U_b - U_c = -V_{S1} + 10 I_1 \Rightarrow I_1 = 2A$$

$$\therefore P = -V_{S1} I_1 = -120W, \text{ 发出功率 } 120W$$

②



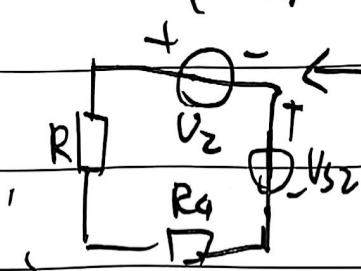
$$U = \frac{R}{R_d + R} U_d$$

$$\left. \begin{array}{l} 5 = \frac{10}{R_d + 10} U_d \\ 8 = \frac{40}{R_d + 40} U_d \end{array} \right\} \Rightarrow \left. \begin{array}{l} R_d = 10 \Omega \\ U_d = 10 V \end{array} \right\}$$

(2) 当 $R = R_d = 10 \Omega$

$$P_{\max} = \left(\frac{U_d}{R + R_d} \right)^2 R = 2.5 W$$

(3).



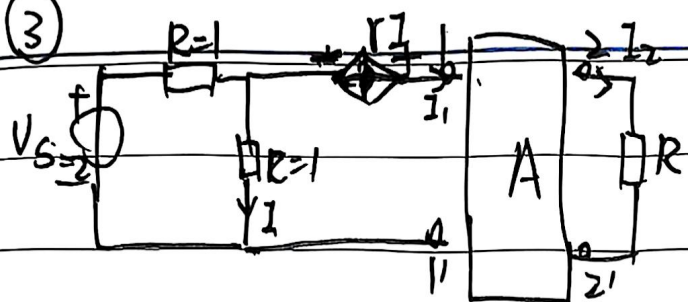
替换: $U_2(U) = \alpha U + b$

$$\left. \begin{array}{l} U_2(5) = 5\alpha + b = 3 \\ U_2(8) = 8\alpha + b = 6 \end{array} \right\} \Rightarrow \left. \begin{array}{l} \alpha = 1 \\ b = -2 \end{array} \right\}$$

$$\therefore U_2 = U - 2$$

$\therefore$ 当 $U = 2V$ 时, $P_{\min} = 0$

③



$$I_2(I_1) = 2I_1 + b$$

$$I_2(0) = b = 0.5A$$

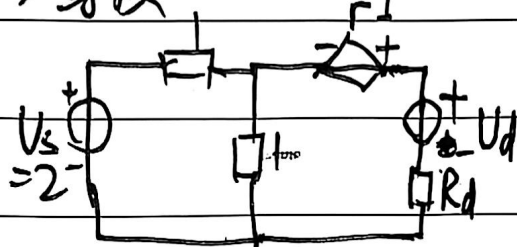
$$I_2\left(\frac{2}{3}\right) = \alpha \cdot \frac{2}{3} + b = 1.5A \Rightarrow$$

$$\begin{cases} b = 0.5 \\ \alpha = 1.5 \end{cases}$$

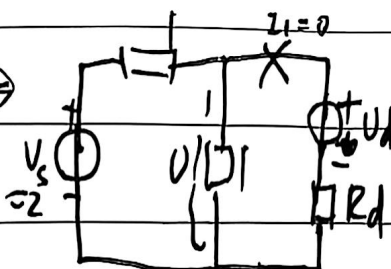
$$\therefore I_2 = \frac{3}{2}I_1 + \frac{1}{2}$$

即求 I_1

$\Rightarrow$ 等效

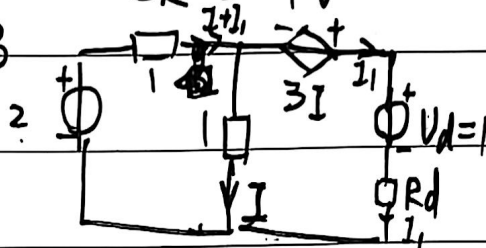


$$1^\circ r = 1 \Rightarrow$$



$$U_d = \frac{R}{2R} V_s = 1V$$

$$2^\circ r = 3$$



$$\begin{cases} 3I = I_1 \\ V_s = 5IR = 1 \\ I_0 = 0.4A \\ I_1 = 1.2 \end{cases}$$

$$\therefore \begin{cases} V_s - IR - (I + I_1)R = 0 \\ (I + I_1)R - 3I + U_d + I_1 R_d - U_s = 0 \end{cases}$$

$$\Rightarrow \begin{cases} I = \frac{2}{3}A \\ R_d = 2.5\Omega \end{cases}$$

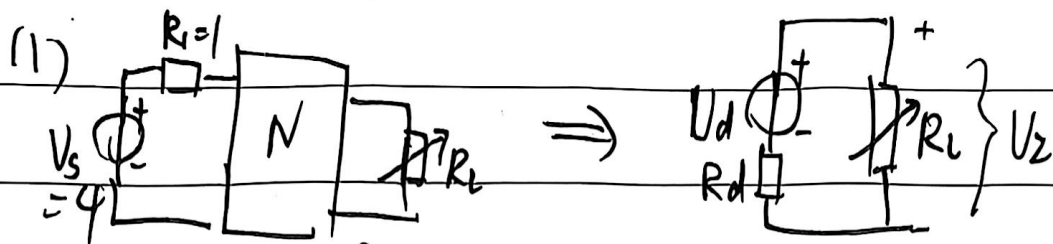
$$R_d = 2.5\Omega$$

$$\therefore r = 5\Omega \text{ 时, } \begin{cases} V_s - IR - (I + I_1)R = 0 \\ (I + I_1)R - 5I + U_d + I_1 R_d - U_s = 0 \end{cases} \Rightarrow$$

$$I_1 = 1A$$

$$\therefore I_2 = 2A$$

④



有:
$$U_2 = \frac{R_L}{R_d + R_L} U_d$$

$$\begin{cases} 6 = \frac{6}{R_d + 6} U_d \\ 9.6 = \frac{12}{R_d + 12} U_d \end{cases} \Rightarrow \begin{cases} U_d = 24V \\ R_d = 18\Omega \end{cases}$$

~~$\therefore$ 当 $R_L = R_d = 18\Omega$ 时, $P_{max} = \left(\frac{U_d}{2R_d}\right)^2 R_L = 8W$~~

②. 当 $U_s = 4V$ 时

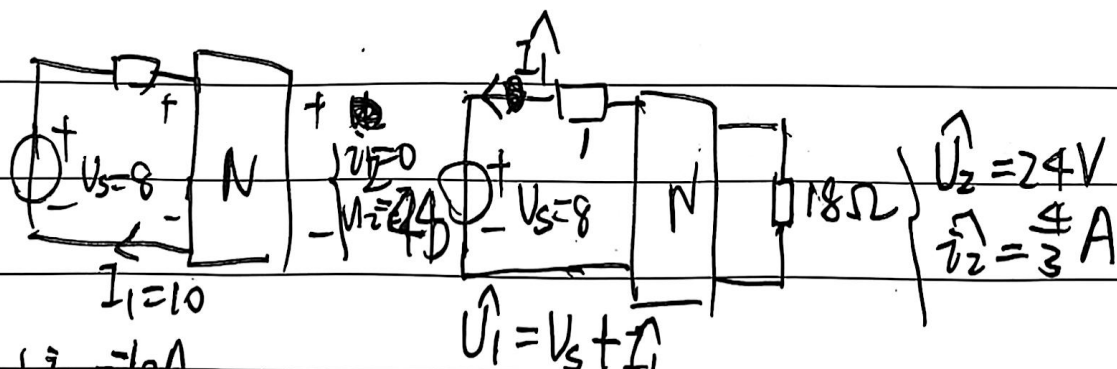
最大功率: $R_L = R_d = 18\Omega$ 时, $P'_{max} = \left(\frac{U_d}{2R_d}\right)^2 R_d = 8W$

$P_{max} = I_2^2 R_L$, $I_2 \propto U_s$ 时 $I_2 = \alpha U_s$

$\therefore P_{max} = \left(\frac{8}{4}\right)^2 P'_{max} = 32W$

(2) $R_L = 18\Omega$

已知 $I_N = 10A$, $I_1 = I_N + I_{R_L}$



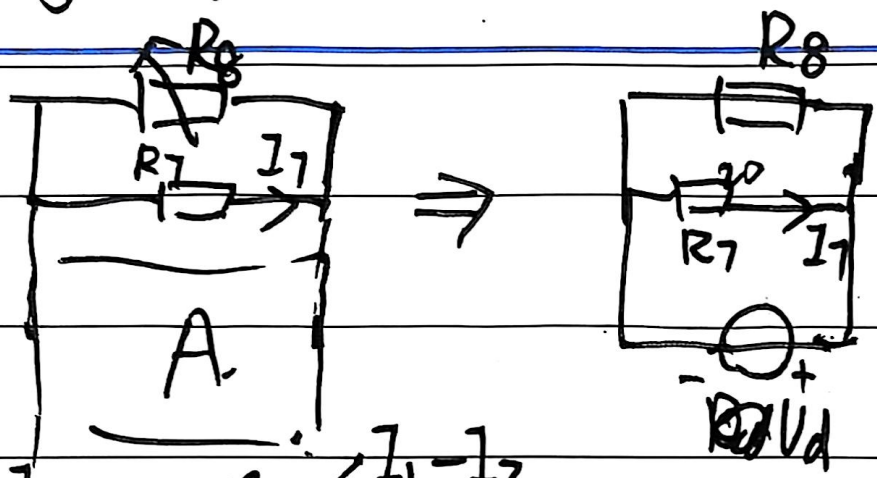
$\begin{cases} I_1 = 10A \\ I_2 = 10A \end{cases}$

$U_1 = U_s - I_1 R_1 = -2V$ $\therefore U_1 \hat{i}_1 + U_2 \hat{i}_2 = \hat{U}_1 \hat{i}_1 + \hat{U}_2 \hat{i}_2$

$\Rightarrow \hat{i}_1 = -18A - 2\hat{i}_1 + 44 \cdot \frac{4}{3} = (8 + \hat{i}_1)(-10)$

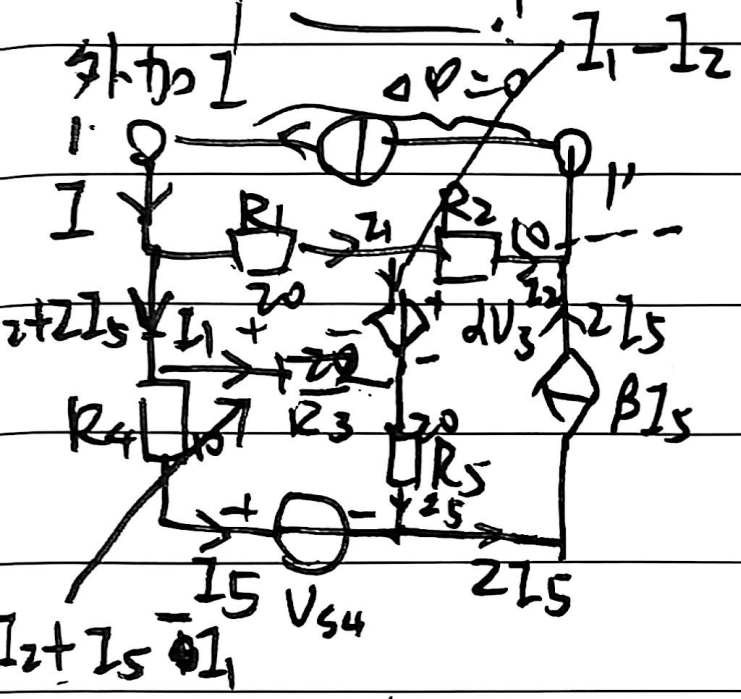
$\therefore P = U_s \cdot \hat{i}_1 = -224W$, $\Rightarrow \hat{i}_1 = -18A$

⑤ $\frac{24}{18} \times \frac{24}{8} = 24$ 16 64 $\frac{18}{18} \times 3$



可知 $R_d = 0$.

I_1, I_2, I_5 及 V_3



$I_1 R_1 + I_2 R_2 = 0$

①: $I_1 R_1 + \alpha V_3 - (I_2 + I_5 - I_1) R_3 = 0$

②: $I_5 R_4 + V_{s4} - I_5 R_5 - (I_2 + I_5 - I_1) R_3 = 0$

③: $\alpha V_3 + I_5 R_5$

$V_3 = (I_2 + I_5 - I_1) R_3$

$\Rightarrow \alpha = 2$